Sets & Venn Diagrams

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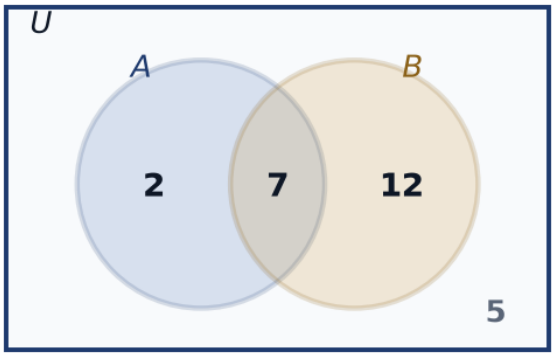
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Transformations and congruence

In the diagram, ABC is a triangle with angle ABC = 90 degrees. AD is the altitude to the hypotenuse BC. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

2 [Foundational] • Target: Congruent triangle tests (Extended)

The diagram shows triangle ABC and its image A'B'C' after a single transformation. Determine the coordinates of A' if A is at $(1, 2)$.



3 [Foundational] • Target: Scale factors

The points A = (2, k), B = (8, 4) and C = (5, 7) form a right-angled triangle with the right angle at B. Find the value of k.

4 [Foundational] • Target: Similar triangle tests

In triangle ABC, $AB = 8$ cm, $BC = 11$ cm, and angle $BAC = 45^\circ$. Find the obtuse angle $\angle ACB$ to 1 decimal place.

5 [Foundational] • Target: Transformations and congruence

In a cyclic quadrilateral $ABCD$, AB is extended to point E. If angle $\angle CBE = 74^\circ$, what is the size of interior angle $\angle ADC$?

6 [Foundational] • Target: Congruent triangle tests (Extended)

In triangle PQR, angle $P = 40^\circ$, angle $Q = 75^\circ$, and side $q = 12$ cm. Use the sine rule to find the length of side p.

7 [Foundational] • Target: Scale factors

In triangle ABC, $a = 5$ cm, $b = 7$ cm, and $c = 8$ cm. Calculate the size of angle B to the nearest degree.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Similar triangle tests

The diagram shows points A, B, and C. Point A' is the image of point A after a translation. Find the coordinates of A'.

9 [Intermediate] • Target: Transformations and congruence

The line segment AB has a midpoint M at $(4, 7)$. Point A is located at $(1, 3)$. Find the coordinates of point B, denoted as (x, y).

10 [Intermediate] • Target: Congruent triangle tests (Extended)

The coordinates of point A are $(-2, 3)$ and the coordinates of point B are $(4, -1)$. Point M is the midpoint of the line segment AB. Find the coordinates of M.

11 [Intermediate] • Target: Scale factors

A solid metal cone with radius 6 cm and height 12 cm is melted down and recast into solid spheres of radius 2 cm. How many complete spheres can be made?

12 [Intermediate] • Target: Similar triangle tests

A bag contains 5 red balls and 3 blue balls. A ball is chosen at random from the bag. What is the probability that the ball chosen is red?

13 [Intermediate] • Target: Transformations and congruence

In the diagram below, lines AB and CD are parallel. A transversal line intersects them at points E and F respectively. If the angle $\text{AEF} = 50^\circ$, find the measure of angle EFC.

14 [Intermediate] • Target: Congruent triangle tests (Extended)

A cuboid has length 10 cm, width 6 cm, and height 4 cm. Calculate the angle that the space diagonal makes with the base of the cuboid.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Scale factors • Cross-Lesson: 7.01 Transformations & Vectors

A bag contains 6 white balls and 4 black balls. Two balls are drawn at random without replacement. What is the probability of drawing one ball of each colour?

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16 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 8.01 Probability & Statistics

On a histogram, the class $0 \leq w < 10$ has a height of 1.4 cm and represents a frequency of 28. What frequency is represented by a bar for class $10 \leq w < 25$ with a height of 2.0 cm?

17 [Challenging] • Target: Transformations and congruence • Cross-Lesson: 9.01 Functions

Ship X sails from port P on a bearing of 040° for 60 miles to point A. Ship Y sails from P on a bearing of 135° for 80 miles to point B. Calculate the distance between A and B to 1 decimal place.

18 [Challenging] • Target: Congruent triangle tests (Extended) • Cross-Lesson: 1.01 Number

A cuboid has a length of USD 8$cm, awidthofUSD5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

19 [Challenging] • Target: Scale factors • Cross-Lesson: 2.01 Algebra & Graphs

The coordinates of point P are $(2, 5)$ and the coordinates of point Q are $(6, 13)$. Calculate the gradient of the line segment PQ.

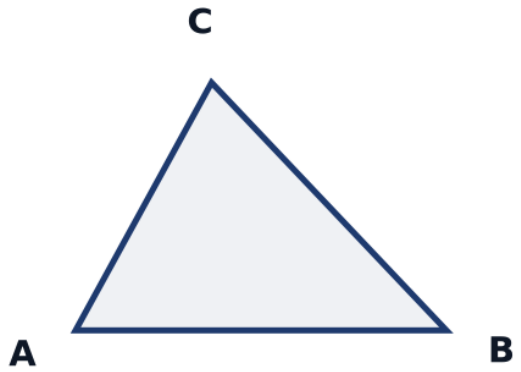
20 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 3.01 Coordinate Geometry

In the diagram, a transversal line intersects two parallel lines. Find the value of x.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Transformations and congruence

In ΔABC studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 103°

b. 113°

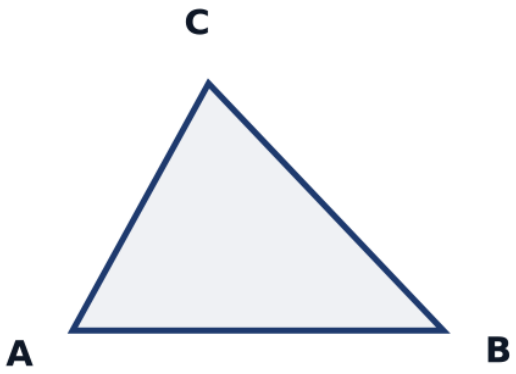
c. 95°

d. 148°

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22 [Foundational] • Target: Transformations and congruence

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

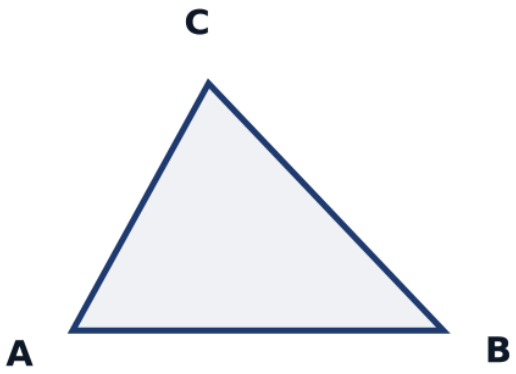
b. 18 cm

c. 24 cm

d. 54 cm

23 [Foundational] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?



a. 9

b. 14

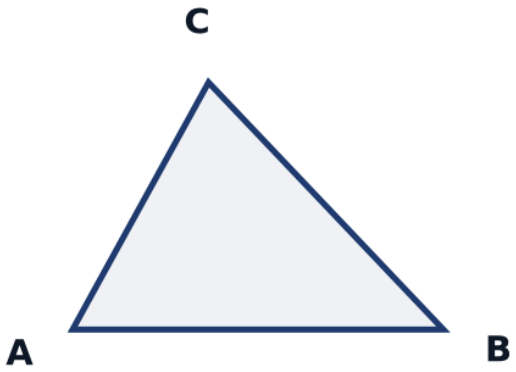
c. 12

d. 15

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24 [Foundational] • Target: Transformations and congruence

In ΔABC studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 82°

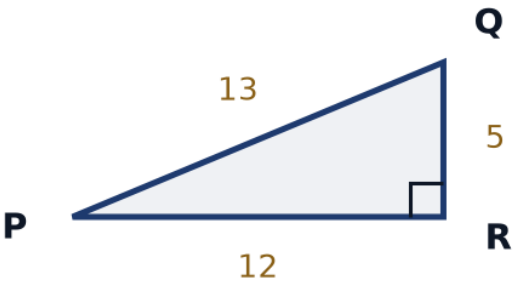
b. 92°

c. 74°

d. 148°

25 [Foundational] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 102°

b. 112°

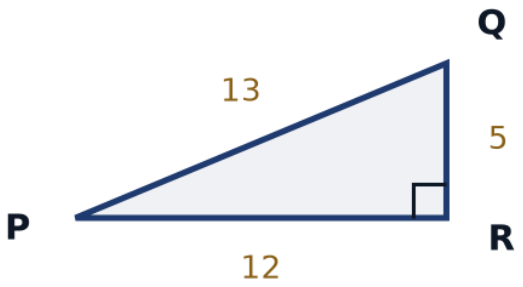
c. 94°

d. 147°

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26 [Foundational] • Target: Congruent triangle tests (Extended)

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?



a. 18 cm

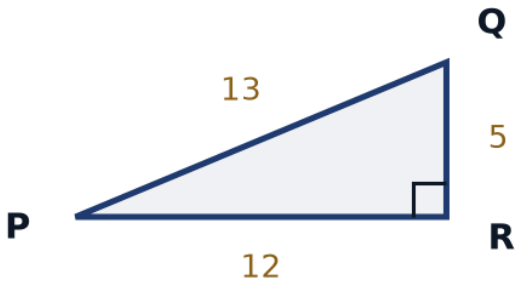
b. 21 cm

c. 28 cm

d. 63 cm

27 [Foundational] • Target: Congruent triangle tests (Extended)

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?



a. 12

b. 18

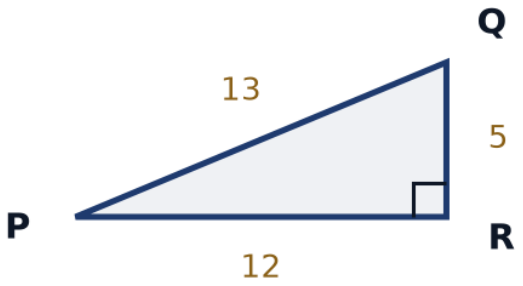
c. 16

d. 20

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28 [Foundational] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 81°

b. 91°

c. 73°

d. 147°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Transformations and congruence

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

30 [Intermediate] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 8

c. 6

d. 9

31 [Intermediate] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 6

b. 11

c. 9

d. 12

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32 [Intermediate] • Target: Transformations and congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

33 [Intermediate] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15

d. 18

34 [Intermediate] • Target: Congruent triangle tests (Extended)

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

35 [Intermediate] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 147°

36 [Intermediate] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 88°

b. 98°

c. 80°

d. 147°

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37 [Intermediate] • Target: Congruent triangle tests (Extended)

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Congruent triangle tests (Extended) • Cross-Lesson: 1.01 Number

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 16

b. 22

c. 20

d. 24

39 [Challenging] • Target: Scale factors • Cross-Lesson: 2.01 Algebra & Graphs

A data set recorded for scale factors consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

40 [Challenging] • Target: Scale factors • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 6x + 5$ in scale factors, evaluate $f(4)$.

a. 29

b. 35

c. 24

d. 33

41 [Challenging] • Target: Scale factors • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

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42 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 5.01 Geometry

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

43 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 6.01 Trigonometry

In ΔABC studying similar triangle tests, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 93°

b. 103°

c. 85°

d. 145°

44 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 7.01 Transformations & Vectors

In ΔABC studying similar triangle tests, $m\angle A = 35^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 86°

b. 96°

c. 78°

d. 145°

45 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 8.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

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